

SC7 Bayes Methods MT25

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Lecture 6: Markov Chain Monte Carlo Methods I

Markov chain Monte Carlo Methods

MCMC is a family of algorithms for simulating $X_0, X_1, X_2, \dots$ so that $X_t \sim p$ (or at least X_t converges to p in distribution) for a user-defined target probability distribution p .

MCMC methods are one of the most versatile classes of Monte Carlo algorithms we have, and are in routine use across statistics.

I will set out the principles for the case that Ω , the space of states of X , is finite (and therefore discrete) and assume they hold for the continuous case.

When we work on a computer we round to computer precision. I call this the **computer measure**. So in practise we always work with a finite set of states.

Markov chains

Let $\{X_t\}_{t=0}^{\infty}$ be a homogeneous Markov chain of random variables on Ω with $X_0 \sim p^{(0)}$ and transition probability

$$P_{i,j} = \mathbb{P}(X_{t+1} = j | X_t = i).$$

Denote by $P_{i,j}^{(n)}$ the n -step transition probabilities

$$P_{i,j}^{(n)} = \mathbb{P}(X_{t+n} = j | X_t = i)$$

and by $p_i^{(t)} = \mathbb{P}(X_t = i)$, with $p^{(t)}$ a row vector.

The transition matrix P is **irreducible** if and only if, for each pair of states $i, j \in \Omega$ there is $n < \infty$ such that $P_{i,j}^{(n)} > 0$. The Markov chain is **aperiodic** if $P_{i,i}^{(n)}$ is non zero for sufficiently large n and all $i \in \Omega$.*

*See Norris, J. *Markov Chains* (2012)

The Stationary Distribution and Detailed Balance

Consider a generic “target” distribution $p_i, i \in \Omega$. This is the distribution we will try to sample. For Bayesian inference, the target distribution will be $\theta \sim \pi(\theta|y)$.

The probability mass function (PMF) $p_i, i \in \Omega, \sum_{i \in \Omega} p_i = 1$ is a stationary distribution of P if $pP = p$. If $p^{(0)} = p$ then

$$p_j^{(1)} = \sum_{i \in \Omega} p_i^{(0)} P_{i,j} = [pP]_j,$$

so $p_j^{(1)} = p_j$ also. Iterating, $p^{(t)} = p$ for each $t = 1, 2, \dots$ in the chain, so the distribution of $X_t \sim p^{(t)}$ doesn't change with t , it is stationary.

We want $X_t \xrightarrow{D} p$. The convergence theorem for finite irreducible Markov chains tells us that if Ω is finite, $pP = p$, and P is irreducible and aperiodic, then indeed $X_t \xrightarrow{D} p$.

Checking $pP = p$ is hard, as we have to sum over all Ω to evaluate pP . However...

Detailed Balance.

If there is a probability mass function $p_i, i \in \Omega$ satisfying $p_i \geq 0$, $\sum_{i \in \Omega} p_i = 1$ and

$$\text{"Detailed balance": } p_i P_{i,j} = p_j P_{j,i} \quad \text{for all } i, j \in \Omega,$$

then $p = pP$ so p is stationary for P .

Detailed balance is sufficient for stationarity, and much easier to check. A Markov chain satisfying DB is **reversible**.

Convergence and the Ergodic Theorem

If $X_t \xrightarrow{D} p$, then $X_t \sim p^{(t)}$ with $p^{(t)} \simeq p$ at large t , so “most” of the samples $X_0, X_1, \dots, X_T$, are “nearly” distributed like p .

For $f : \Omega \rightarrow R$ let $\hat{f}_T = T^{-1} \sum_t f(X_t)$ estimate $E_{X \sim p}(f(X))$.

Theorem. If $\{X_t\}_{t=0}^\infty$ is an irreducible and aperiodic Markov chain on a finite space of states Ω satisfying detailed balance with respect to the probability distribution p , then as $T \rightarrow \infty$

$$P(X_T = i) \rightarrow p_i \text{ and } \hat{f}_T \xrightarrow{a.s.} E_{X \sim p}(f(X))$$

for any bounded function $f : \Omega \rightarrow R$.

Such a chain is **ergodic** with target distribution $p_i, i \in \Omega$.

Metropolis-Hastings Algorithm

Target $p_i, i \in \Omega$ with Ω a finite set. The algorithm simulates X_{t+1} given X_t . This determines $P(X_{t+1} = j | X_t = i)$ and P .

If $X_t = i$ simulate X_{t+1} by accepting or rejecting $j \sim q(\cdot|i)$. This is called the “proposal distribution”.

Let $X_t = i$. X_{t+1} is determined in the following way.

[1] Draw $j \sim q(\cdot|i)$ and $u \sim U[0, 1]$.

[2] If $u \leq \alpha(j|i)$ where

$$\alpha(j|i) = \min \left\{ 1, \frac{p_j q(i|j)}{p_i q(j|i)} \right\}$$

then set $X_{t+1} = j$, otherwise set $X_{t+1} = i$.

We initialise this with some $X_0 = i_0, p_{i_0} > 0$ and iterate for $t = 1, 2, 3, \dots, T$ to simulate the samples we need.

Example: Simulating the hypergeometric distribution

HyperGeom($k; K, N, n$) with $K = 10, N = 20, n = 10$ gives the probability for k successes in n draws from a population of size N containing K successes.

If $p_k = \text{HyperGeom}(k; K, N, n)$ then for $k \in \Omega$,

$$\Omega = \{k \in \mathbb{Z} : \max\{0, n + K - N\} \leq k \leq \min\{n, K\}\}$$

Let $B^- = \max\{0, n + K - N\}$ and $B^+ = \min\{n, K\}$.*.

The PMF is

$$p_k = \binom{K}{k} \binom{N - K}{n - k} / \binom{N}{n}, \quad k \in \Omega$$

Give a MH MCMC algorithm ergodic for p .

*In our example $B^- = 0$ and $B^+ = 10$

Step 1: Choose a proposal distribution $q(j|i)$ that's easy to simulate and irreducible. A simple distribution that 'will do' is

$$q(j|i) = \begin{cases} 1/2 & \text{for } j = i \pm 1 \\ 0 & \text{otherwise,} \end{cases}$$

It's easy to simulate and irreducible: we can get from any state $i \in \Omega$ to any other state $j \in \Omega$ by adding or subtracting 1's.

Notice we can leave the state space $\Omega = \{B^-, B^- + 1, \dots, B^+\}$ given above. If $i = B^+$ and we propose $j = i + 1$ then j has zero probability in the target distribution.

One transparent way to deal with this is give these states probability zero in the target, setting $p_j = 0$ for all $j \notin \Omega$.

Step 2: write down the algorithm.

If $X_t = i$, then X_{t+1} is determined in the following way.

[1] Simulate $j \sim U\{i-1, i+1\}$ and $u \sim U[0, 1]$.

[2] If $B^- \leq j \leq B^+$ and

$$\begin{aligned} u &\leq \min \left\{ 1, \frac{p_j q(i|j)}{p_i q(j|i)} \right\} \\ &= \min \left\{ 1, \frac{\binom{K}{j} \binom{N-K}{n-j}}{\binom{K}{i} \binom{N-K}{n-i}} \right\} \end{aligned}$$

then set $X_{t+1} = j$, else (if either condition fails) set $X_{t+1} = i$.

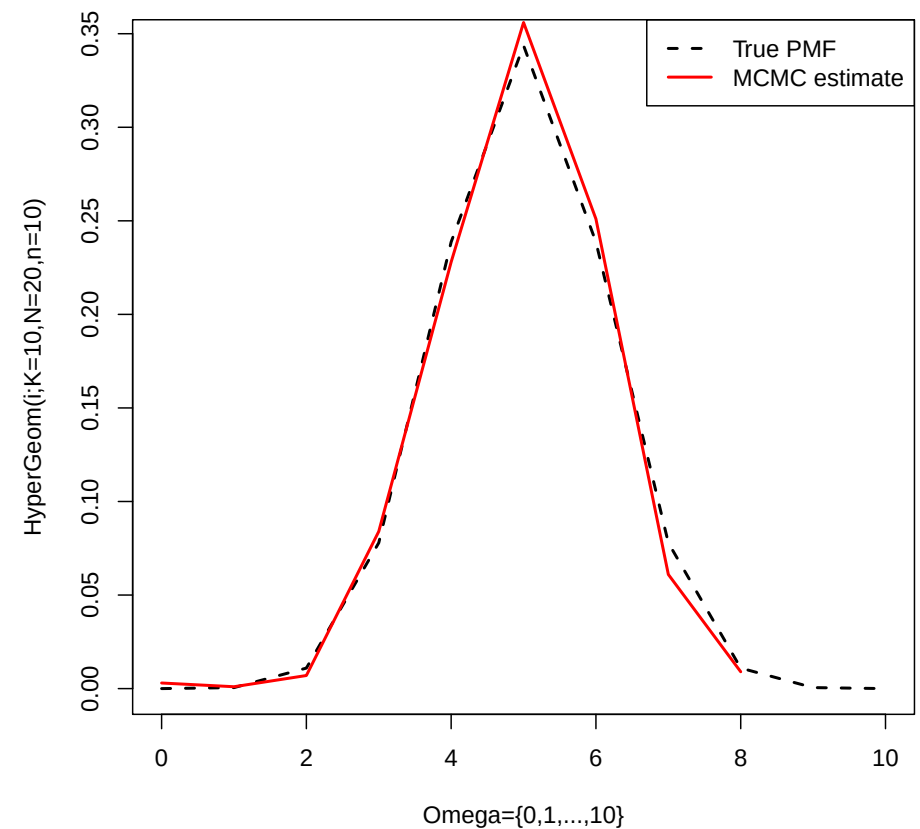
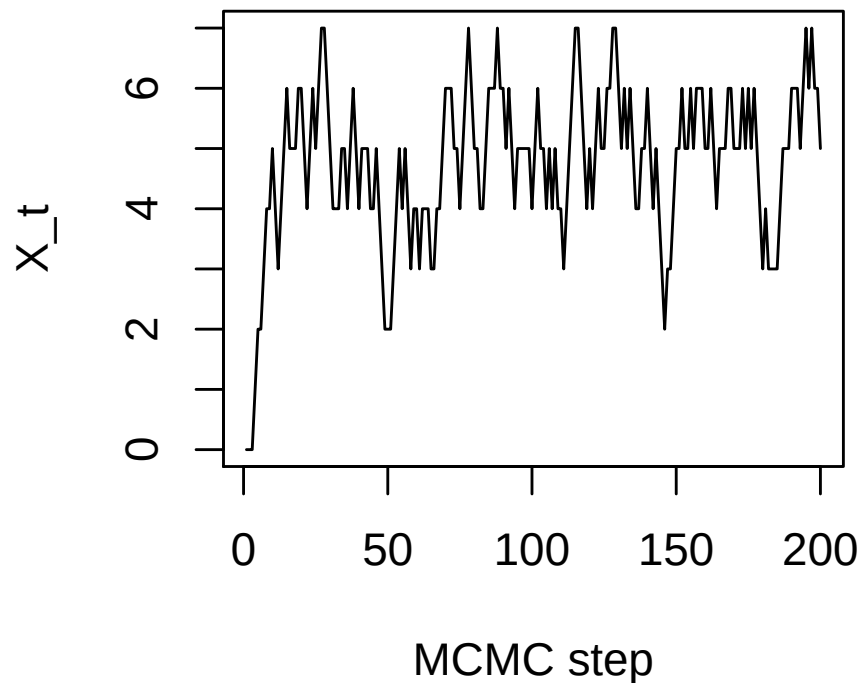
If $j < B^-$ or $j > B^+$ then $p_j = 0$ so $\alpha = 0$. Reject, stay in Ω .

Step 3: check chain is irreducible and aperiodic. Clear here.

Left: x -axis is step counter $t = 1, 2, 3 \dots 200$. The y -axis is Markov chain state X_t targeting

HyperGeom($K = 10, N = 20, n = 10$).

Right: histogram of $X_1, X_2, \dots, X_n$ for $T = 1000$.



Ergodicity proof for Metropolis Hastings

Assume the chain is irreducible and aperiodic and show the chain determined by the MCMC update has stationary distribution p .

Compute the transition matrix P and verify detailed balance,

$$P_{i,j} p_i = P_{j,i} p_j,$$

since that implies stationarity, $p = pP$. DB for $i = j$ is trivial.

Suppose $j \neq i$. If $X_t = i$, then the probability $P_{i,j}$ to move to $X_{t+1} = j$ at the next step is

$$P_{i,j} = P(X_{t+1} = j | X_t = i) = q(j|i) \alpha(j|i).$$

Now check DB:

$$\begin{aligned} p_i P_{i,j} &= p_i q(j|i) \alpha(j|i) \\ &= p_i q(j|i) \min \left\{ 1, \frac{p_j q(i|j)}{p_i q(j|i)} \right\} \\ &= \min \{ p_i q(j|i), p_j q(i|j) \} \\ &= p_j q(i|j) \min \left\{ \frac{p_i q(j|i)}{p_j q(i|j)}, 1 \right\} \\ &= p_i q(i|j) \alpha(i|j) \\ &= p_j P_{j,i} \end{aligned}$$

and we are done.

MCMC for continuous distributions

The candidate state $\theta' \sim q(\theta'|\theta)$ has distribution $q(d\theta'|\theta)$.

MCMC targeting $p(\theta)$, $\theta \in \Omega$: Let $X_t = \theta$.

[1] Draw $\theta' \sim q(\cdot|\theta)$ and $u \sim U[0, 1]$.

[2] If $u \leq \alpha(\theta'|\theta)$ where

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{p(\theta') q(\theta|\theta')}{p(\theta) q(\theta'|\theta)} \right\}$$

then set $X_{t+1} = \theta'$, otherwise set $X_{t+1} = \theta$.

The rejection probability when the current state is $X_t = \theta$ is

$$c(\theta) = 1 - \int_{\Omega} \alpha(\theta'|\theta) q(d\theta'|\theta)$$

The transition distribution $K(\theta, A) = \mathbb{P}(X_{t+1} \in A | X_t = \theta)$ is

$$K(\theta, d\theta') = \alpha(\theta'|\theta) q(d\theta'|\theta) + c(\theta)\delta_\theta(d\theta').$$

The delta function $\delta_\theta(d\theta')$ satisfies $\int_A \delta_\theta(d\theta') = \mathbb{I}_{\theta \in A}$.

When $\theta' \neq \theta$ the transition distribution has a simple unnormalised density

$$K(\theta, \theta') = \alpha(\theta'|\theta) q(\theta'|\theta).$$

Detailed balance becomes

$$K(\theta, \theta') p(\theta) = K(\theta', \theta) p(\theta').$$

Exercise: verify DB for K given above. Similar to discrete case.

MH example: an equal mixture of bivariate normals

$$p(\theta) = (2\pi)^{-1} \left(0.5e^{-(\theta-\mu_1)^T \Sigma_1^{-1} (\theta-\mu_1)/2} + 0.5e^{-(\theta-\mu_2)^T \Sigma_2^{-1} (\theta-\mu_2)/2} \right)$$

with $\theta = (\theta_1, \theta_2)$. Use $\mu_1 = (1, 1)^T$, $\mu_2 = (4, 4)^T$ and $\Sigma_1 = \Sigma_2 = I_2$ for this illustration.

Step 1. For a proposal distribution q we want something simple to sample. A simple choice that “will do” is

$$\theta'_i \sim U(\theta_i - a, \theta_i + a), \quad i = 1, 2$$

with $a > 0$ a fixed constant. We jump uniformly in a box of side $2a$, easy to sample, and

$$q(\theta'|\theta) = q(\theta|\theta') = 1/4a^2.$$

Step 2. The algorithm is, given $\theta^{(t)} = (\theta_1, \theta_2)$,

[1] for $i = 1, 2$ simulate $\theta'_i \sim U(\theta_i - a, \theta_i + a)$

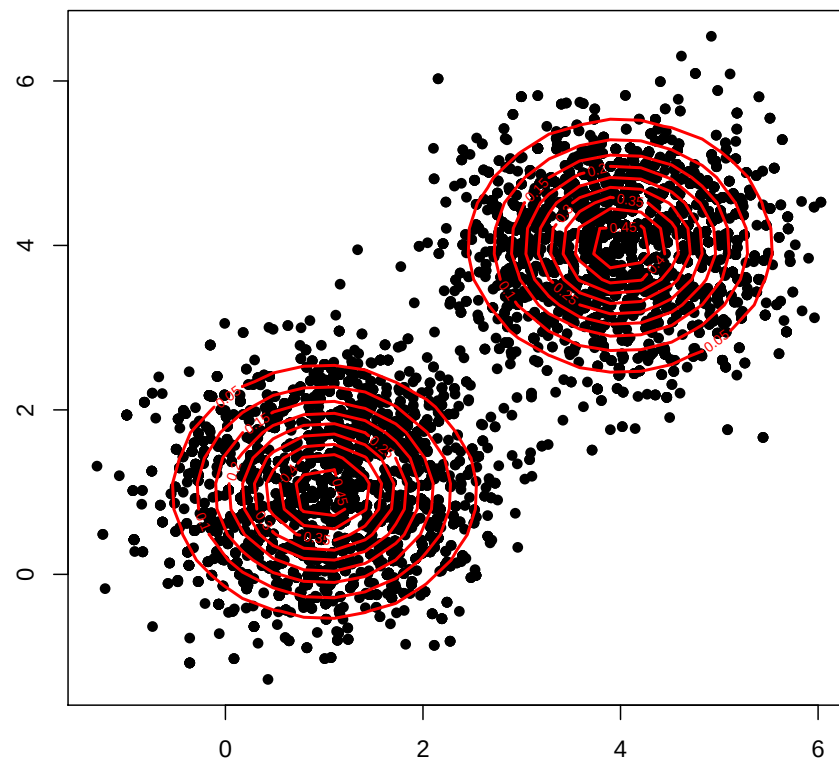
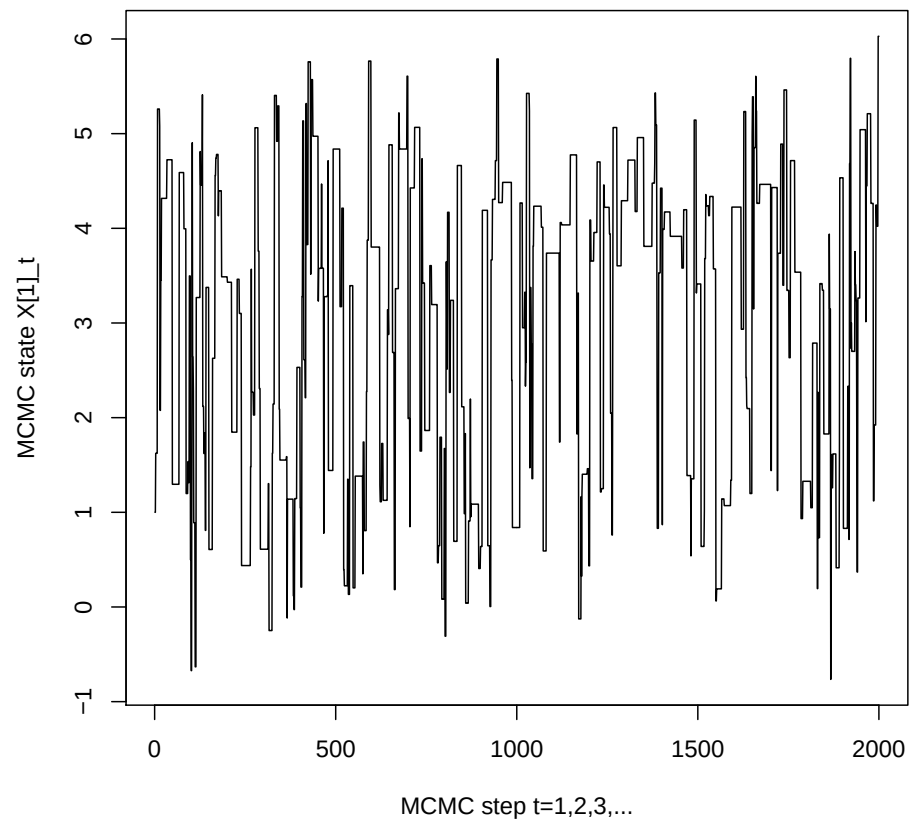
[2] with probability

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{p(\theta')}{p(\theta)} \right\}$$

set $\theta^{(t+1)} = \theta'$ otherwise set $\theta^{(t+1)} = \theta$.

Step 3. This algorithm is ergodic for any $a > 0$ but we will see that the choice of a makes a difference to efficiency.

Exercise: (see the associated R-file) try running this, experimenting with different values of a .



Mixing updates for multivariate targets

If $\theta = (\theta_1, \dots, \theta_p)$ and we target $p(\theta)$, $\theta \in \mathbb{R}^p$, update one variable at a time. Small changes have higher acceptance probability.

Choose a component $i \sim \xi$ with probability ξ_i , $i = 1, \dots, p$, set $\theta'_{-i} = \theta_{-i}$ and $\theta'_i \sim q_i(\cdot|\theta)$. The acceptance probability is

$$\alpha_i(\theta'|\theta) = \min \left\{ 1, \frac{p(\theta') q_i(\theta|\theta')}{p(\theta) q_i(\theta'|\theta)} \right\}$$

and the (unnormalised) transition density for $\theta' \neq \theta$ is

$$K(\theta, \theta') = \sum_i \xi_i K_i(\theta, \theta'),$$

where $K_i = q_i(\theta'_i|\theta) \alpha_i(\theta'|\theta)$. A mixture of updates satisfying DB also satisfies DB so the chain targets $p(\theta)$.

Example: normal mixture one variable at a time

Let $\theta^{(t)} = (\theta_1, \theta_2)$. Take $\xi = (1/2, 1/2)$ and $q_i(\theta'_i | \theta_{-i}) = 1/2a$.

1. Simulate $i \sim U\{1, 2\}$, $\theta'_i \sim U(\theta_i - a, \theta_i + a)$ and set $\theta'_{-i} = \theta_{-i}$.
2. With probability

$$\alpha_i(\theta' | \theta) = \min \left\{ 1, p(\theta') / p(\theta) \right\}$$

set $\theta^{(t+1)} = \theta'$, otherwise set $\theta^{(t+1)} = \theta$.

